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Birth Cohort Study of Early Life Exposure to Air Pollutants in Beijing

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Birth Cohort Study of Early Life Exposure to Air Pollutants in Beijing

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ABSTRACT

Introduction: Association between early exposure to ambient air pollution and adverse pregnancy outcomes is well-established in high-income Western countries. However, the matter is less clear in China. This study will assess the risk of early life exposure to air pollutants in Beijing and explore the significance of using 8-hydroxy-deoxyguanosine as a biological indicator to assess early life exposure to environmental pollution. Research into genomics and metabolomics studies will also be carried out to investigate the mechanisms underlying adverse pregnancy outcomes caused by environmental factors in order to reduce maternal and infant morbidity and mortality. **Methods and analysis:** For this study, 2000 women with singleton pregnancies and their infants will be recruited and studied to establish a birth cohort from Jan. 2017 to Dec. 2018 in Beijing Obstetrics and Gynecology Hospital at Capital Medical University. The infants will be followed up for 12 months after birth. Biological sample collection will be performed based on the maternal and birth cohort. A total of 50,000 samples will be collected during different trimesters, and 10 biological sample libraries will be established for maternal serum, plasma, urine, placental tissue, umbilical cord tissue, and umbilical cord blood. The level of environmental pollution and population exposure will be evaluated using environmental monitoring data published daily by the Beijing Municipal Environmental Monitoring Station near the pregnant women's residences. A database of maternal and newborn information and biological sample library based on the cohort will be established to study the correlations

between early life environmental exposure and adverse pregnancy outcomes. Maternal and umbilical cord blood samples (1000 each) will be randomly selected from the cohort for 8-hydroxy-deoxyguanosine assay to determine the correlation between early life exposures and oxidative stress in pregnant women and fetuses. Cases with adverse outcomes will be selected and compared to those of healthy pregnant women and newborns to analyze the mechanisms underlying adverse pregnancy outcomes caused by environmental factors. SPSS version 18.0 (SPSS, Chicago, IL, U.S.) and SAS version 9.3 (SAS Institute Inc., Cary, NC, U.S.) statistical software will be used for data analysis.

Ethics and dissemination: This trial will be conducted in accordance with the Declaration of Helsinki and supervised by the Medical Ethics Committee of Beijing Obstetrics and Gynecology Hospital, Capital Medical University (approval number 2016-KY-059-01). All patients will receive information about the study and written forms and will be provided informed consent before enrolment. The findings of this study will be disseminated via the usual scientific forums including peer reviewed publication and presentation at international conferences.

Strengths and limitations

1. This is the first birth cohort study focusing on the early life exposure to air pollutants and adverse pregnancy outcomes in Beijing.
2. This research will aim to further investigate the mechanisms underlying the impact of air pollution to adverse birth outcomes.
3. This study should provide a reference for future large-scale studies of early life exposure to air pollutants in China.
4. Restriction to a single city limits the generalize-ability of our research, because the proportion of PM2.5 from regional and local sources varies by geographic location.
5. In this study, we take account for age, socioeconomic predictors, daily activities and time spent indoors as part of the confounding, however, individual level exposure not quantified

by personal sampling. Quantification of an individual's exposure is imprecise without either a completely controlled environment or portable sampling equipment.

INTRODUCTION

Maternal and child health is the premise and basis of sustainable human development, it is also related to the happiness and well-being of millions of households. Thus, people are making every effort to reduce the incidence of adverse pregnancy outcomes and neonatal morbidity. The rapid economic development and urbanization in China has led to worsening trends in the overall urban and regional environmental qualities [1,2]. The impact of environmental pollution on human health has long attracted attention. Environmental pollution not only directly causes elevated population mortality and increased morbidity related to respiratory and cardiovascular diseases, but also leads to the elevated incidence of adverse pregnancy outcomes [3-6]. In addition, it affects infants and young children and also produces long-term effects on adult health, including inducing hypertension, coronary heart disease, and non-insulin-dependent diabetes [7]. The impact of environmental pollution on adverse pregnancy outcomes has been studied internationally for some time, and the epidemiological studies include cross-sectional surveys, follow-ups and cohort studies [8-10], which have strong abilities to demonstrate a causal relationship. Ecological studies such as time series and case-crossovers have been used to study the impact of air pollution on adverse pregnancy outcomes so as to analyze the impact of air pollution (PM_{2.5}, PM₁₀, SO₂, NO₂, O₃) on low birth weight, premature birth, birth defects, stillbirth, intrauterine inflammation and mothers with preeclampsia or diabetes mellitus [11-15], and epidemiological evidence has been presented [16] In recent years, with the development of technology and cross-disciplinary approaches, estimation of individual maternal pollution exposure is achieved by combining data from environmental monitoring stations and residential

addresses. An increasing number of studies have examined the effects of air pollution on adverse pregnancy outcomes, including studies in China [17,18]. Compared with Western countries, China has relatively high levels of air pollution levels to the World Health Organization air pollution database. However, the research on the effect of environmental pollution on adverse pregnancy outcomes in China has lagged and is quite limited. Environmental pollution studies have mainly focused on persistent organic pollutants, heavy metal pollution and atmospheric pollution. Most studies used multivariate logistic regression or time series to analyze the relationship between the exposure to atmospheric PM10, SO2, NO2 during pregnancy and adverse pregnancy outcomes. The main adverse pregnancy outcomes studied include preterm birth, birth defects and low birth weight. The national environmental protection bureau of China monitored PM2.5 in 2013. There are only a few studies directly linking PM2.5 to long-term health effects in China. Given the large number of births in China, the impact of environmental pollution on adverse pregnancy outcomes cannot be neglected. However, studies on this issue in China are relatively limited. Thus, it is necessary to conduct large-sample prospective cohort studies to quantitatively evaluate individual exposure and to analyze the impact of environmental pollution on birth outcomes. In addition, these studies allow further investigation of the interaction between the environment and genetics by integrating epidemiological and biological mechanisms.

OBJECTIVES

Primary Objectives

The primary aims of this study are to establish a maternal and birth cohort, then set up a maternal and newborn information database and biological sample library based on the cohort.

Collect the air pollutant indicators published daily by the Environmental Monitoring Station near the pregnant women's residences, including PM_{2.5}, PM₁₀, SO₂, and NO at different stages of pregnancy.

To study the correlations between early life environmental exposures and adverse pregnancy outcomes, and to investigate the correlations between different levels of exposure to ambient air pollution at different pregnancy stages and adverse pregnancy outcomes including preterm birth, fetal growth restriction, low birth weight, stillbirth, as well as the correlations to birth defects like congenital heart disease, neural tube defects, cleft lip and palate, and reproductive abnormalities.

Secondary Objectives

To explore the significance of using 8-hydroxy-deoxyguanosine as a biological indicator to detect early life exposures to environmental pollutions.

Carry out a nested control study for cases with adverse pregnancy outcomes and birth defects to investigate the mechanisms underlying these adverse outcomes caused by environmental factors, so as to prevent adverse outcomes caused by early life exposures.

STUDY DESIGN

Study Population and Cohort Design

There is a prospective cohort study. A total of 2000 singleton pregnant women will be recruited. From Jan. 1, 2017 to Dec. 31, 2018, pregnant women will be recruited during their routine antenatal check at the Beijing Obstetrics and Gynecology Hospital of Capital Medical University; they will be delivering in the hospital and planning to reside in Beijing for at least 3

years. A total of 2000 singleton pregnancy women who are willing to participate and provide written informed consent will be recruited, they will also agree that their newborns will participate in the birth cohort for a 1-year follow up. This study is performed to establish a maternal and birth cohort and a maternal and newborn information database and biological sample library based on this cohort for the purpose of studying the correlations between early life environmental exposure and adverse pregnancy outcomes. A flow chart of this trail is given in Figure 1.

Exclusion Criteria

History of radioactive material contact during early pregnancy, history of using teratogenic drugs or fever during early pregnancy (body temperature >38.5°C); family history of genetic disease; inbreeding; chromosome abnormalities of pregnant women or their husbands.

Data Collection and Management

1. Data collection of participant

Pregnant women will be recruited according to the criteria. After written informed consents are provided, they will fill out the questionnaires with the help of trained investigators. Maternal population economics information, detailed residential address three months before pregnancy and during early pregnancy, health condition, lifestyle, history of pregnancy, anamnesis, mode of conception, and condition during early pregnancy will be noted. Detailed individual information collected from pregnant women (through a questionnaire survey, detailed in Questionnaires): Age, height, weight, body mass index, occupation (occupational exposure), education and other demographic data. Detailed residential address, daily outdoor activities and time, outdoor

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3 exertion and protective measures during smog episodes, usage of air purifying apparatus, and
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5 cooking frequencies. Health condition before pregnancy, menstrual history, anamnesis and
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7 pregnancy history, mode of conception, history of smoking or passive smoking, alcohol
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9 consumption, lifestyle during three months before pregnancy, history of adverse environmental
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11 exposures, dietary habits, folic acid usage, morning sickness, mental health, and some
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13 information on the husband, including height, weight, occupation, smoking, and alcohol
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15 assumption. Each individual's information will be entered into the database and numbered, and
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17 her infant will be assigned a matched number. The pregnant women will enter the cohort at early
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19 pregnancy and be followed through the whole process of early pregnancy (6–13+6 gestational
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21 weeks), the second trimester (14–27+6 weeks), the third trimester (> 28 weeks), at birth, 42 days
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23 after birth, neonatal and infant stage; information will be systematically collected, including
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25 clinical data such as clinical phenotype, laboratory results, ultrasound images, treatments and
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27 prognosis, as well as growth data of the infant. Newborns will enter the birth cohort upon birth.
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29 A medical database with 2000 pairs of mother and infant covering the whole process through
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31 pregnancy to newborn until infant will be established. Outcome measurements will be check by a
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33 clinical expert. These data will be written on the case report form(CRF) by a certificated clinical
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35 research coordinator.
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43 **2. Biological sample collection**

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45 Biological sample collection will be initiated based on the establishment of the maternal and
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47 birth cohort. A total of 50000 samples are expected, and 10 biological sample libraries will be
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49 established, which will contain maternal serum, plasma, urine, placental tissue, umbilical cord
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51 tissue, umbilical cord blood.
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Maternal peripheral venous blood (5 ml) will be collected during early pregnancy, the second trimester and third trimester, and 24 hours after birth, plasma or serum will be separated and cryopreserved at -80°C . Urine (5 ml) will be cryopreserved. Amniotic fluid samples (10 ml) will be collected in cases who receive induced abortion due to fetal malformations found in the second trimester.

Umbilical cord blood (5 ml), umbilical cord (10 cm) and $2\times 2\times 2$ cm full-thickness placenta samples (rinsed in ice-cold saline to remove blood) will be collected and cryopreserved at -80°C . 5 ml of maternal urine and a strand of hair 3 mm in length will also collected and cryopreserved in the library. The delivery process and newborn's condition will be recorded in detail, and the survey questionnaire will be completed by the physician. A detailed birth index will be recorded, which will include gestational week, gender, weight, length, Apgar score, neonatal complications, and appearance and organ abnormalities.

3. Infancy visits

An early growth monitoring system will be established based on the maternal and birth cohort and started from the birth of the newborns. Anthropometric measurements and questionnaires related to feeding and sickness will be conducted at 42 days, 6 and 12 months of age in a community health care center. 42 days after birth, the mother will revisit with the newborn. Information pertaining to infant growth and development, as well as breastfeeding will be recorded, and the survey questionnaire will be completed by the physician. At 6 and 12 months after birth, the infant will be brought to the community health care clinic for anthropometric measurements and development evaluation; infancy development will be monitored, the height, weight, head circumference and number of teeth will be measured, and

questionnaires will be completed regarding feeding and illness information including diseases of the respiratory or cardiovascular system diagnosed by doctors.

4. Air pollution data collection

The environmental pollution level and population exposure will be evaluated using environmental monitoring data (PM_{2.5}, PM₁₀, SO₂, NO, and O₃). Data (PM_{2.5}, PM₁₀, SO₂, NO, and O₃) published daily by the Beijing Municipal Environmental Monitoring Station near the pregnant women's residences will be collected from Oct 1, 2016 to Oct 31, 2019. Daily pollution data of ultrafine particles and ozone will be monitored by the research group itself by establishing monitoring instruments in the communities where the study subjects live; daily meteorological data such as temperature and humidity will also be collected and used as a confounding control in the analysis.

5. Recording of outcome measures

Adverse pregnancy outcomes include birth defects, abortion, preterm birth, low birth weight, fetal growth restriction, stillbirth, and fetal retardation. Among these, the birth defects include anencephaly, neural tube defects (anencephaly, spina bifida, congenital spina bifida, congenital hydrocephalus), cleft lip and palate (cleft palate, cleft lip, cleft lip with cleft palate), ear deformity (microtia including anotia, outer ear and other malformations), gastrointestinal abnormalities (esophageal atresia or stenosis, anorectal atresia or stenosis (including no anus), urinary tract malformations (hypospadias, bladder exstrophy), abdominal malformation (congenital diaphragmatic hernia, omphalocele, gastroschisis), Down syndrome (trisomy 21), congenital heart disease, limb malformations (varus clubfoot, polydactyly, syndactyly and longitudinal deficiency), limb shortening, and ambiguous genitalia. Preterm birth refers to birth at a gestational age of 28 to 36⁺⁶ weeks (gestational age is calculated according to the last

menstrual period combined with an early pregnancy ultrasound image). Low birth weight refers to birth weight below 2500 g. Abortion refers to termination of pregnancy before 28 weeks of gestational age or fetal weight below 1000 g. Induced abortion is excluded.

Data Analysis

1. Investigate the relationship between adverse environmental exposures of the population and offspring development outcomes (that is, adverse pregnancy outcomes)

Data will be analyzed to evaluate the association between environmental exposures and pregnancy outcomes. Then a narrative description of all available evidence is provided. All study participants will be closely followed up throughout the pregnancy, and pregnancy complications (such as pregnancy-induced hypertension, gestational diabetes, fetal growth restriction, premature rupture of membranes, and intrauterine infection) will be recorded in detail in a timely manner and considered confounding factors during statistical analysis.

2. Investigate the relationship between early life exposure and oxidative stress in pregnant women and fetuses

Maternal and umbilical cord blood samples (1000 each) will be randomly selected from the cohort for 8-hydroxy-deoxyguanosine assay to determine the correlation of early life exposures to oxidative stress in pregnant women and fetuses.

3. Further nested case control study

Cases with adverse outcomes including birth defects, preterm birth, stillbirth, and low birth weight will be selected and compared with healthy pregnant women and newborns to analyze the correlation of maternal early exposures with adverse pregnancy outcomes and offspring

developmental abnormalities. Genomics and metabolomics studies will also be carried out to investigate the mechanisms of adverse pregnancy outcomes caused by environmental factors.

Quality Control

Data management: The questionnaires were designed based on extensive literature reviews and advice from obstetrics and pediatrics experts. Presurveys have been conducted at Beijing Obstetrics and Gynecology Hospital, and the questionnaires will be revised and modified before they are finalized. Detailed standards will be established for subject recruitment of the maternal and birth cohort. Inclusion and exclusion criteria will be specified, and participants will be closely monitored through the entire pregnancy and birth until 12 months after birth to ensure that the recorded information is accurate and reliable. The exact residence information of the pregnant women will be used to determine where to set up the environmental monitoring stations; the environmental data will be recorded daily to ensure accuracy. All privacy sensitive data will be removed and all identifying information of participants will be coded. Returned coded and anonymized questionnaires will be stored in a locked separately from study records that link to participant identifying codes. Data monitoring center is located in Beijing Obstetrics and Gynecology Hospital, Scientific research department is responsible for data management. DMC is independent from the sponsor and have no competing interests. The participants may drop out of the trial at any time for any reason. The physician should perform assessment and give the proper assistance. Incidences of loss to follow-up will be reported to the chief designer of this trial every week.

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Investigator training: A research manual (CRF tables) will be prepared to standardize the investigation and survey techniques and to improve the working ability and responsibility of investigators; specialized physicians will be responsible for the clinical diagnostic reports.

Questionnaire verification: Clinical physicians will check the questionnaires every time after the survey for logic errors and missed information, and additional surveys will be conducted within 24 hours to supplement the data.

Data entry: A combination of hard copy questionnaires and computer database entries will be used for data collection. Concurrent validity data will be collected for quality control. Training will be provided to all personnel involved in data entry beforehand. Two specialized personnel will be responsible for the second round of quality control assessment every month.

Biological sample collection protocol development: Sample collection will be performed in strictly accordance with protocol, and the lost-to-follow-up rate will be closely controlled. Biological sample libraries will be managed by assigned individuals, SOP regulations will be developed, and sample collection location, size, standing time, transportation, and storage will be strictly controlled according to the protocol. To ensure data accuracy, completeness and clarity, all samples will be stored using the sample number matching the maternal information and recorded and managed via computer management software system, which will be docked with the hospital HIS system and regularly exported for storage.

Adverse events

Either pregnancy women or infants', any adverse event should be recorded in detail and closely monitored before stabilization or resolution. The chief designer of this trial will be informed immediately and responsible for reporting to Scientific research department and

Medical Ethics Committee in Beijing Obstetrics and Gynecology Hospital. Regular monitoring will be used to ensure that all adverse events are identified and addressed appropriately.

Statistical Analysis

Epidate 3.1 software will be used for double-entry of the data and logic verification to establish a database; SPSS version 18.0 (SPSS, Chicago, IL, U.S.) and SAS version 9.3 (SAS Institute Inc., Cary, NC, U.S.) statistical software will be used for data analysis. Descriptive analysis will be performed first to compute the general maternal and infant demographic data and distribution of birth outcomes and birth defects. Assessment methods pertaining to personal exposure to the main air pollutants will be established by using Beijing road traffic data and land use data to build a land-usage regression model or time-spatial distribution pattern estimation; the variables included are elevation, population density, industrial land, green space and traffic. Cox regression and competing risk analysis will be used to compute the hazard ratio (HR) and population attributable risk (PAR) and to estimate the effect of main air pollutants on major adverse birth outcomes (low birth weight, large for gestational age and fetal growth restriction) and birth defects (congenital heart disease).

DISCUSSION

Epidemiological studies in Europe and the United States have shown that exposure to air pollution during pregnancy is associated with adverse pregnancy outcomes [19–21]. With China's rapid development, air pollution has become more serious than before, especially in Beijing [22]. The monthly average PM_{2.5} concentrations is 127.1 µg/m³ in December 2016 in Beijing, which is higher than the current U.S. EPA National Ambient Air Quality Standard (NAAQS) of 12 µg/m³. However, only a few studies have directly linked PM_{2.5} to pregnancy

outcomes in China. It is not clear whether premature birth, low birth weight, stillbirth, congenital anomalies, other adverse maternal outcomes, or infant respiratory or circulatory issues are significantly associated with environment exposure. This study will establish a birth cohort in Beijing and investigate whether there are any associations between prenatal exposure to air pollution and adverse pregnancy outcomes. However, this work will be explored the mechanisms underlying the impact of early life exposures to adverse pregnancy outcomes in order to decrease the occurrence of birth defects and adverse pregnancy outcomes due to environmental factors, reduce infant respiratory and cardiovascular diseases, and improve population quality.

Ethical Approval

The protocol for recruitment and collection of baseline data and biological sample for the cohort has be approved by the Medical Ethics Committee of Beijing Obstetrics and Gynecology Hospital, Capital Medical University.

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Contributors

YC is the chief designer of this trial. JS, LW, YM, NT, LHZ, YYK and YMD are coinvestigators. YMD assisted in designing the study and developed the objectives and methods of the study. JS and LW drafted the manuscript, YC is responsible for supervision of the study and revision of the manuscript. All authors have contributed to the writing of the study protocol and have approved the final version of this manuscript. YC, JS, LW, YM, NT are involved in the

recruitment of patients. LHZ are responsible for the management of biological sample library.

YYK is responsible for neonatal follow-up.

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JS: Dr. Jing Song

LW: Dr. Ling Wei

YM: Dr. Ying Ma

NT: Dr. Ning Tian

YMD: Prof. Yin Mei Dai

LHZ: Dr. Li Hong Zhao

YYK: Dr. Yuan Yuan Kong

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Data Sharing Statement

The findings of this study will be disseminated via the usual scientific forums including peer reviewed publication and presentation at international conferences.

Competing Interests

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None declared.

Trail Registration Number

This study has been registered in WHO International Clinical Trial register- Chinese Clinical Trial Registry with registration number ChiCTR-ROC-16010181.
(<http://www.chictr.org.cn/showproj.aspx?proj=17328>)

For peer review only

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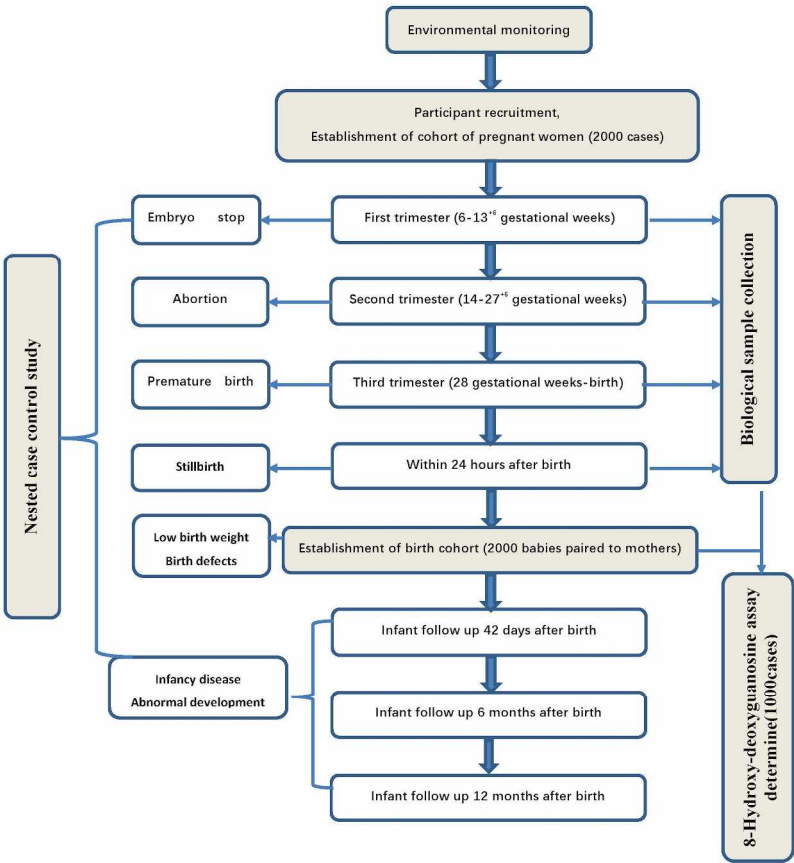
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Figure 1: Flow chart



224x290mm (600 x 600 DPI)

Section/Item	Item number	Description	
Administrative information			
Title	1	Descriptive title identifying the study design, population, interventions, and, if applicable, trial acronym	1
Trial registration	2a	Trial identifier and registry name. If not yet registered, name of intended registry.	16
	2b	All items from the World Health Organization Trial Registration Data Set	n.a.
Protocol version	3	Date and version identifier	2
Funding	4	Sources and types of financial, material, and other support	16
Roles and responsibilities	5a	Names, affiliations, and roles of protocol contributors	15
	5b	Name and contact information for the trial sponsor	15
	5c	Role of study sponsor and funders, if any, in study design; collection, management, analysis, and interpretation of data; writing of the report; and the decision to submit the report for publication, including whether they will have ultimate authority over any of these activities	16
	5d	Composition, roles, and responsibilities of the coordinating center, steering committee, end point adjudication committee, data management team, and other individuals or groups overseeing the trial, if applicable (see item 21a for DMC)	12
Introduction			
Background and rationale	6a	Description of research question and justification for undertaking the trial, including summary of relevant studies (published and unpublished) examining benefits and harms for each intervention	4
	6b	Explanation for choice of comparators	n.a.
Objectives	7	Specific objectives or hypotheses	5-6
Trial design	8	Description of trial design, including type of trial (e.g., parallel group, crossover, factorial, single group), allocation ratio, and framework (e.g., superiority, equivalence, noninferiority, exploratory)	6
Methods			
Participants, interventions, and outcomes			
Study setting	9	Description of study settings (e.g., community clinic, academic hospital) and list of countries where data will be collected. Reference to where list of study sites can be obtained	7-10
Eligibility criteria	10	Inclusion and exclusion criteria for participants. If applicable, eligibility criteria for study centers and individuals who will perform the interventions (e.g., surgeons, psychotherapists)	7
Interventions	11a	Interventions for each group with sufficient detail to allow replication, including how and when they will be administered	n.a.
	11b	Criteria for discontinuing or modifying allocated interventions for a given trial participant (e.g., drug dose change in response to harms, participant request, or improving/worsening disease)	12
	11c	Strategies to improve adherence to intervention protocols, and any procedures for monitoring adherence (e.g., drug tablet return, laboratory tests)	n.a.

Section/Item	Item Number	Description	
	11d	Relevant concomitant care and interventions that are permitted or prohibited during the trial	n.a.
Outcomes	12	Primary, secondary, and other outcomes, including the specific measurement variable (e.g., systolic blood pressure), analysis metric (e.g., change from baseline, final value, time to event), method of aggregation (e.g., median, proportion), and time point for each outcome. Explanation of the clinical relevance of chosen efficacy and harm outcomes is strongly recommended	10
Participant timeline	13	Time schedule of enrollment, interventions (including any runins and washouts), assessments, and visits for participants. A schematic diagram is highly recommended (Figure).	7-9
Sample size	14	Estimated number of participants needed to achieve study objectives and how it was determined, including clinical and statistical assumptions supporting any sample size calculations	6
Recruitment	15	Strategies for achieving adequate participant enrollment to reach target sample size	7
Assignment of interventions (for controlled trials)			
Allocation Sequence generation	16a	Method of generating the allocation sequence (e.g., computer-generated random numbers), and list of any factors for stratification. To reduce predictability of a random sequence, details of any planned restriction (e.g., blocking) should be provided in a separate document that is unavailable to those who enroll participants or assign interventions.	n.a.
Allocation concealment mechanism	16b	Mechanism of implementing the allocation sequence (e.g., central telephone; sequentially numbered, opaque, sealed envelopes), describing any steps to conceal the sequence until interventions are assigned	n.a.
Implementation	16c	Who will generate the allocation sequence, who will enroll participants, and who will assign participants to interventions	n.a.
Blinding (masking)	17a	Who will be blinded after assignment to interventions (e.g., trial participants, care providers, outcome assessors, data analysts), and how	12
	17b	If blinded, circumstances under which unblinding is permissible, and procedure for revealing a participant's allocated intervention during the trial	n.a.
Data collection, management, and analysis			
Data collection methods	18a	Plans for assessment and collection of outcome, baseline, and other trial data, including any related processes to promote data quality (e.g., duplicate measurements, training of assessors) and a description of study instruments (e.g., questionnaires, laboratory tests) along with their reliability and validity, if known. Reference to where data collection forms can be found, if not in the protocol.	7-10
	18b	Plans to promote participant retention and complete follow-up, including list of any outcome data to be collected for participants who discontinue or deviate from intervention protocols	12
Data management	19	Plans for data entry, coding, security, and storage, including any related processes to promote data quality (e.g., double data entry; range checks for data values). Reference to where details of data management procedures can be found, if not in the protocol.	11-13
Statistical methods	20a	Statistical methods for analyzing primary and secondary outcomes. Reference to where other details of the statistical analysis plan can be found, if not in the protocol.	13-14
	20b	Methods for any additional analyses (e.g., subgroup and adjusted analyses)	n.a.
	20c	Definition of analysis population relating to protocol nonadherence (e.g., as- randomized analysis), and any statistical methods to handle missing data (e.g., multiple imputation)	n.a.

Monitoring			
Data monitoring	21a	Composition of DMC; summary of its role and reporting structure; statement of whether it is independent from the sponsor and competing interests; and reference to where further details about its charter can be found, if not in the protocol. Alternatively, an explanation of why a DMC is not needed.	12
	21b	Description of any interim analyses and stopping guidelines, including who will have access to these interim results and make the final decision to terminate the trial	12
Harms	22	Plans for collecting, assessing, reporting, and managing solicited and spontaneously reported adverse events and other unintended effects of trial interventions or trial conduct	13
Auditing	23	Frequency and procedures for auditing trial conduct, if any, and whether the process will be independent from investigators and the sponsor	n.a.
Ethics and dissemination			
Research ethics approval	24	Plans for seeking REC/IRB approval	15
Protocol amendments	25	Plans for communicating important protocol modifications (e.g., changes to eligibility criteria, outcomes, analyses) to relevant parties (e.g., investigators, RECs/IRBs, trial participants, trial registries, journals, regulators)	n.a.
Consent or assent	26a	Who will obtain informed consent or assent from potential trial participants or authorized surrogates, and how (see item 32)	7
	26b	Additional consent provisions for collection and use of participant data and biological specimens in ancillary studies, if applicable	8-9
Confidentiality	27	How personal information about potential and enrolled participants will be collected, shared, and maintained in order to protect confidentiality before, during, and after the trial	12
Declaration of interests	28	Financial and other competing interests for principal investigators for the overall trial and each study site	16
Access to data	29	Statement of who will have access to the final trial data set, and disclosure of contractual agreements that limit such access for investigators	16
Ancillary and post-trial care	30	Provisions, if any, for ancillary and post-trial care, and for compensation to those who suffer harm from trial participation	n.a.
Dissemination policy	31a	Plans for investigators and sponsor to communicate trial results to participants, health care professionals, the public, and other relevant groups (e.g., via publication, reporting in results databases, or other data-sharing arrangements), including any publication restrictions	16
	31b	Authorship eligibility guidelines and any intended use of professional writers	n.a.
	31c	Plans, if any, for granting public access to the full protocol, participant-level data set, and statistical code	n.a.
Appendices			
Informed consent materials	32	Model consent form and other related documentation given to participants and authorized surrogates	20
Biological specimens	33	Plans for collection, laboratory evaluation, and storage of biological specimens for genetic or molecular analysis in the current trial and for future use in ancillary studies, if applicable	13

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Early Life Exposure to Air Pollutants and adverse pregnancy outcomes: survey protocol in Beijing

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Manuscripts

**Early Life Exposure to Air Pollutants and adverse pregnancy outcomes:
survey protocol in Beijing**

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Key Word: Early life exposure, air pollutants, birth cohort, adverse pregnancy outcomes, 8-hydroxy-deoxyguanosine

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ABSTRACT

Introduction: The association between early exposure to ambient air pollution and adverse pregnancy outcomes is less clear in China. This study will assess the risk of early life exposure to air pollutants in Beijing and explore the viability of 8-hydroxy-deoxyguanosine as a biological indicator to assess oxidative stress induced by early life exposure to air pollution.

Methods and analysis: 2000 women with singleton pregnancies and their infants will be recruited from the Beijing Obstetrics and Gynecology Hospital to establish birth cohort. We will collect 10 types of biological samples, including maternal serum, plasma, urine, placental tissue, umbilical cord tissue, and umbilical cord blood during all three trimesters. The air pollution data (PM2.5, PM10, and so on) will be record at official fixed-site monitoring stations closest to where the pregnant women live. We will assess the effect of air pollutants on adverse pregnancy outcomes, and infant respiratory and circulatory disease using Cox regression and competitive risk analysis and explore possible critical windows of exposure during pregnancy using daily pollutant concentrations averaged over various period of pregnancy combined with the individual activity patterns and physiological parameters. Maternal and umbilical cord blood

samples (1000 each) will be randomly selected for 8-hydroxy-deoxyguanosine assay to assess the correlation between air pollutants exposures and oxidative stress. We will investigate whether air pollutant exposure or 8-OHdG levels are associated with adverse pregnancy outcomes. SPSS and SAS statistical software will be used for data analysis. Cox regression and competing risk analysis will be used to compute the hazard ratio and population attributable risk .

Ethics and dissemination: This research is approved by the Medical Ethics Committee of Beijing Obstetrics and Gynecology Hospita. Written informed consent will be obtained from all study participants prior to enrollment. The results will be published in peer-reviewed journals or disseminated through conference presentations.

Strengths and limitations

1. This is the first birth cohort study to focus on the early-life exposure to air pollutants and adverse pregnancy outcomes in Beijing.
2. This research aims to further investigate the mechanisms underlying the impact of air pollution to adverse birth outcomes.
3. This study should provide a reference for future large-scale studies of early life exposure to air pollutants in China.
4. Restriction to a single city limits the generalizability of our research, because the proportion of PM2.5 from regional and local sources varies by geographic location.
5. In this study, we consider age, socioeconomic predictors, daily activities and time spent indoors confounding factors. However, individual-level exposure will not be quantified by personal sampling. Quantification of an individual’ s exposure is imprecise without either a completely controlled environment or portable sampling equipment.

INTRODUCTION

Maternal and child health is the premise and basis of sustainable human development; it is also related to the happiness and well-being of millions of households. Thus, people are making every effort to reduce the incidence of adverse pregnancy outcomes and neonatal morbidity. The rapid economic development and urbanization of China have led to worsening trends in the overall quality of urban and regional environments [1,2]. The impact of air pollution on human health has long attracted attention. Air pollution is not only directly linked to elevated mortality and morbidity through respiratory and cardiovascular diseases, but also leads to adverse pregnancy outcomes [3-6]. In addition, it not only affects infants directly, but it can also have a long-term impact on the health of adults, including inducing hypertension, coronary heart disease, and non-insulin-dependent diabetes [7]. The impact of air pollution on adverse pregnancy outcomes has been studied internationally for some time, and the epidemiological studies include cross-sectional surveys, follow-ups and cohort studies [8-10], which have strong abilities to demonstrate a causal relationship. Time series and case-crossovers have been used to study the impact of air pollution (PM_{2.5}, PM₁₀, SO₂, NO₂ and O₃) on adverse pregnancy outcomes such as low birth weight, premature birth, birth defects, stillbirth, and intrauterine inflammation, and maternal conditions such as preeclampsia and diabetes mellitus [11-16]. In recent years, with the development of technology and cross-disciplinary approaches, estimation of individual maternal exposure to air pollutants is achieved by combining data from environmental monitoring stations and residential addresses. An increasing number of studies have examined the effects of air pollution on adverse pregnancy outcomes, including studies in China [17,18]. According to the World Health Organization air pollution database, China has higher levels of air pollution than Western countries. However, the research into the effect of air pollution on adverse pregnancy outcomes in China has lagged and is quite limited. Environmental pollution studies have focused

mainly on persistent organic pollutants, heavy metal pollution and atmospheric pollution. Most studies have used multivariate logistic regression or time series to analyze the relationship between the exposure to atmospheric PM10, SO₂, and NO₂ during pregnancy and adverse pregnancy outcomes. The main adverse pregnancy outcomes studied have been preterm birth, birth defects and low birth weight. The national environmental protection bureau of China monitored PM2.5 in 2013. There are only a few studies directly linking PM2.5 to long-term health effects in China. Given the large number of births in China, the impact of air pollution on adverse pregnancy outcomes must not be disregarded. However, studies on this issue in China are relatively limited. Thus, it is necessary to conduct large-sample prospective cohort studies to quantitatively evaluate individual exposure and to analyze the impact of air pollution on birth outcomes. In addition, these studies allow further investigation of the interaction between the air pollutant exposure and genetics by integrating epidemiological and biological mechanisms.

OBJECTIVES

Primary Objectives

The primary aims of this study are to establish a maternal and birth cohort, then set up a maternal and newborn information database and biological sample library based on that cohort.

We will record the air pollutant indicators published daily by the Environmental Monitoring Station near the pregnant women’s residences, including PM2.5, PM10, SO₂, and NO₂ at different stages of pregnancy.

To study the correlations between environmental exposures early in life and adverse pregnancy outcomes, and to investigate the correlations between different levels of exposure to

1
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3 ambient air pollution at different stages of pregnancy and adverse pregnancy outcomes including
4 preterm birth, fetal growth restriction, low birth weight, and stillbirth and the correlations to birth
5 defects like congenital heart disease, neural tube defects, cleft lip and palate, and reproductive
6 abnormalities.
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15 **Secondary Objectives**

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17 To explore the significance of using 8-hydroxy-deoxyguanosine as a biological indicator to
18 assess early life exposures to air pollutants.
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22 To perform a nested control study for cases with adverse pregnancy outcomes and birth
23 defects and investigate the mechanisms underlying these adverse outcomes caused by air
24 pollutant factors and so prevent adverse outcomes caused by early life exposures.
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32 **STUDY DESIGN**

33 **Study Population and Cohort Design**

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35 This is a prospective cohort study. A total of 2000 singleton pregnant women will be
36 recruited from Jan. 1, 2017 to Dec. 31, 2018 during their routine antenatal check at the Beijing
37 Obstetrics and Gynecology Hospital of Capital Medical University. They will undergo delivering
38 in the hospital and must plan to reside in Beijing for at least 3 years. A total of 2000 women with
39 singleton pregnancies who are willing to participate and provide written informed consent will
40 be recruited. They must also agree for their newborns to participate in the birth cohort for a
41 1-year follow up. This research is performed to establish a maternal and birth cohort and
42 biological sample library studying covering correlations between exposure to air pollutants early
43 in life and adverse pregnancy outcomes. A flow chart of this trail is given in Figure 1.
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Exclusion Criteria

Each patient’s history of radioactive material contact during early pregnancy, history of use of any teratogenic drugs or fever during early pregnancy (body temperature >38.5°C); family history of genetic disease; inbreeding; and chromosome abnormalities of pregnant women or their husbands are to be recorded.

Data Collection and Management

1. collection of participant data

Pregnant women will be recruited according to the inclusion criteria. After providing written informed consent, they will fill out the questionnaires with the help of trained investigators. The maternal population’s financial information, detailed residential history going back three months before pregnancy and during early pregnancy, health condition, lifestyle, history of pregnancy, anamnesis, mode of conception, and condition during early pregnancy will be noted. Detailed individual information collected from pregnant women (through questionnaire , detailed in Questionnaires): Age, height, weight, body mass index, occupation (occupational exposure), education and other demographic data. Detailed residential address, daily outdoor activities and time spent outdoors, outdoor exertion and protective measures used during smog episodes, usage of air purifying apparatus, and cooking habits. Health condition before pregnancy, menstrual history, anamnesis and pregnancy history, mode of conception, history of smoking or passive smoking, alcohol consumption, lifestyle during three months before pregnancy, history of adverse environmental exposure, dietary habits, folic acid usage, morning sickness, mental health, and some information on the husband, including height, weight,

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3 occupation, smoking, and alcohol assumption will be recorded. Everyone's information will be
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5 entered into the database and numbered, and each infant will be assigned a number matching his
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7 or her mother's. The pregnant women will enter the cohort early during pregnancy and be
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9 followed throughout early pregnancy (6–13⁺⁶ gestational weeks), the second trimester (14–27⁺⁶
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11 weeks), the third trimester (> 28 weeks), at birth, 42 days after birth, and the neonatal and infant
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13 stages; information including clinical data such as clinical phenotype, laboratory results,
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15 ultrasound images, treatments and prognosis, and infant growth data will be systematically
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17 collected. Newborns will enter the birth cohort upon birth. A medical database with 2000 pairs
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19 of mothers and infants covering the entire process through pregnancy and infancy will be
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21 established. Outcome measurements will be checked by a clinical expert. These data will be written
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23 on the case report form (CRF) by a certificated clinical research coordinator.
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29 **2. Biological sample collection**

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31 Biological sample collection will be initiated based on the establishment of the maternal and
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33 birth cohort. Biological sample libraries containing maternal serum, plasma, and urine collected
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35 during the 1st trimester (1–13⁺⁶ weeks), 2nd trimester (14–27⁺⁶ weeks), 3rd trimester (28 weeks to
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37 delivery), and at birth will be established. Placental tissue, umbilical cord tissue, umbilical cord
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39 blood, meconium, children's dried blood spot and maternal hair and colostrum/breast milk will
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41 be collected after delivery. CVS samples and amniotic fluid samples (10 ml) will be collected from
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43 cases who receive prenatal diagnosis of fetal malformations found. A total of more than 38000
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45 samples are expected.
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51 Maternal peripheral venous blood (5 ml) and urine (5 ml) will be collected during early
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53 pregnancy, the second trimester, the third trimester, and 24 h after birth. Plasma or serum and
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55 urine will be separated and cryopreserved at –80°C. CVS samples and amniotic fluid samples
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(10 ml) will be collected from cases who received a prenatal diagnosis due to fetal malformations found in the second trimester.

Umbilical cord blood (5 ml), umbilical cord (10 cm) and 2×2×2 cm full-thickness placental samples (rinsed in ice-cold saline to remove blood) will be collected and cryopreserved at –80°C at birth. Then 5 ml of maternal urine, a strand of hair 3 mm in length and maternal colostrum/breast milk (1 ml), will be collected 24 hours after birth and cryopreserved in the library. The delivery process and newborn’s condition will be recorded in detail, and the survey questionnaire will be completed by the physician. A detailed birth index will be recorded. It will include gestational week, infant gender, weight, length, Apgar score, neonatal complications, and appearance and organ abnormalities. Each baby’s meconium will also be collect to examine metabolites and assess the fecal microbiota. Dried blood spot specimens in one heel of the newborn will be collect onto specially manufactured absorbent filter paper and store in plastic bag.

3. Infancy visits

An early growth monitoring system will be established based on the maternal and birth cohort and newborns will be included starting at birth. Anthropometric measurements and questionnaires related to feeding and sickness will be conducted at a community health care center when the children are 42 days, 6 months, and 12 months old. At 42 days after birth, the mother will bring the newborn to the health center. Information pertaining to infant growth and development and breastfeeding will be recorded, and the survey questionnaire will be completed by the physician. At 6 and 12 months after birth, the infant will be brought to the community health care clinic for anthropometric measurements and evaluation of its development. Infant development will be monitored, and height, weight, head circumference and number of teeth will

be determined, and questionnaires will be completed regarding feeding and illness information, including diseases of the respiratory or cardiovascular system as diagnosed by doctors.

4. Air pollution data collection for exposure assessment

There are 35 air quality monitoring stations distributed in the different districts in Beijing which is in line with national air quality monitoring position with the requirement of technical specification. The air pollution level and population exposure will be evaluated using environmental monitoring data (PM_{2.5}, PM₁₀, SO₂, NO₂, and O₃) published daily by the Beijing Municipal Environmental Monitoring Center. Data come from the closest monitoring station where pregnant women live. Daily meteorological data such as temperature and humidity will also be collected from the national meteorological center and used as a confounding control in the analysis. We explore potential critical windows of exposure during pregnancy using daily PM_{2.5}, PM₁₀ and other pollutant concentration averaged over various period including the three months before pregnancy and each of the three trimesters of pregnancy. Exposure throughout the entire pregnancy will be calculated as the mean of all days of the pregnancy.

5. Recording of outcome measures

Adverse pregnancy outcomes include birth defects, abortion, preterm birth, low birth weight, fetal growth restriction, stillbirth, and fetal retardation. Among these, the birth defects include anencephaly, neural tube defects (anencephaly, spina bifida, congenital spina bifida, congenital hydrocephalus), cleft lip and palate (cleft palate, cleft lip, cleft lip with cleft palate), ear deformity (microtia including anotia, outer ear and other malformations), gastrointestinal abnormalities (esophageal atresia or stenosis, anorectal atresia or stenosis (including no anus), urinary tract malformations (hypospadias, bladder exstrophy), abdominal malformation (congenital diaphragmatic hernia, omphalocele, gastroschisis), Down syndrome (trisomy 21),

congenital heart disease, limb malformations (varus clubfoot, polydactyly, syndactyly and longitudinal deficiency), limb shortening, and ambiguous genitalia. Preterm birth refers to birth at a gestational age of 28 to 36⁺⁶ weeks (gestational age is calculated according to the last menstrual period combined with an early pregnancy ultrasound image). Low birth weight refers to birth weight below 2500 g. Abortion refers to termination of pregnancy before 28 weeks of gestational age.

Data Analysis

1. Investigate the relationship between adverse air pollutant exposure of the population and offspring development outcomes (specifically adverse pregnancy outcomes)

Data will be analyzed to evaluate the association between air pollutant exposures and pregnancy outcomes. All study participants will be closely followed up throughout the pregnancy, and pregnancy complications (such as pregnancy-induced hypertension, gestational diabetes, fetal growth restriction, premature rupture of membranes, and intrauterine infection) will be recorded in detail in a timely manner. Based on PM2.5, PM10, SO₂, NO₂, and O₃ concentrations and many kinds of calculations methods such as the closest point method and inverse distance weighting, combining with the individual activity patterns and physiological parameters, hazard ratio and population attributable risk will be calculated using Cox regression and competitive risk analysis in order to assess the effect of air pollutants for adverse pregnancy outcomes.

2. Investigate the relationship between early life exposure and oxidative stress in pregnant women and fetuses

Some studies showed that exposure to particulate air pollution early in life plays a role in increasing systemic oxidative stress. Oxidative stress is believed to be involved in the mechanisms underlying the effects of air pollution on human health. 8-hydroxy-2'-deoxyguanosine (8-OHdG) is a sensitive indicator of oxidative stress. We will investigate 8-OHdG levels during different time windows in maternal blood and cord blood and explore the relationship between exposure to air pollutants and oxidative stress. Maternal and umbilical cord blood samples (1000 each) will be randomly selected from the cohort for 8-hydroxy-deoxyguanosine assay to determine the correlation of early life exposures to oxidative stress in pregnant women and fetuses. All recruited pregnant women paired with their babies will be numbered starting at 1, and then either odd or even will be randomly selected for testing.

3. Further nested case control study

Cases with adverse outcomes including birth defects, preterm births, stillbirths, and low birth weight will be selected and compared to healthy pregnant women and newborns to analyze the correlation of maternal early exposures with adverse pregnancy outcomes and offspring developmental abnormalities. Genomic and metabolomic experiments will also be carried out to investigate the mechanisms of adverse pregnancy outcomes caused by environmental factors.

Quality Control

Data management: The questionnaires were designed based on extensive literature reviews and advice from obstetrics and pediatrics experts. Detailed standards will be established for subject recruitment of the maternal and birth cohort. The exact residence information of the pregnant women will be recorded to allow selection of the closest environmental monitoring stations. Each data monitoring center is located in Beijing Obstetrics and Gynecology Hospital. The participants may drop out of the trial at any time for any reason. The physician should assess

each patient and provide any needed assistance. Incidences of loss to follow-up will be reported to the chief designer of this trial every weekly.

Investigator training: A research manual (CRF tables) will be prepared to standardize the investigation and survey techniques and to improve the working ability and responsibility of investigators; specialized physicians will be responsible for the clinical diagnostic reports.

Questionnaire verification: Clinical physicians will check the questionnaires after every survey for logical errors and missed information, and additional surveys will be conducted within 24 h to supplement the data.

Data entry: A combination of hard copy questionnaires and computer database entries will be used for data collection. Concurrent validity data will be collected for quality control. Training will be provided to all personnel involved in data entry beforehand. Two specialized personnel will be responsible for the second round of quality control assessment every month.

Development of a protocol for biological sample collection : Sample collection will be performed in strict accordance with protocol, and the rate of loss to follow-up will be closely controlled. Biological sample libraries will be managed by assigned individuals, SOP regulations will be developed, and sample collection site size, standing time, transportation, and storage will be strictly controlled according to the protocol. To ensure data accuracy, completeness and clarity, all samples will be stored using the sample number matching the maternal information and recorded and managed via computer management software system, which will be docked with the hospital HIS system and regularly exported for storage.

Adverse events

adverse event in any pregnant woman or infant will be recorded in detail and closely monitored before stabilization or resolution. The chief designer of this trial will be informed

immediately and responsible for reporting to Scientific Research Department and Medical Ethics Committee of Beijing Obstetrics and Gynecology Hospital. Regular monitoring will be used to ensure that all adverse events are identified and addressed appropriately.

Statistical Analysis

Epidata 3.1 software will be used for double-entry of the data and logic verification to establish a database; SPSS version 18.0 (SPSS, Chicago, IL, U.S.) and SAS version 9.3 (SAS Institute Inc., Cary, NC, U.S.) statistical software will be used for data analysis. Descriptive analysis will be performed first to compute the general maternal and infant demographic data and distribution of birth outcomes and birth defects. Cox regression and competing risk analysis will be used to compute the hazard ratio (HR) and population attributable risk (PAR) and to estimate the effect of main air pollutants on major adverse birth outcomes (low birth weight, large for gestational age and fetal growth restriction) and birth defects (congenital heart disease). Pearson correlation coefficients and multiple linear regression will be used to assess the association between air pollutants exposure and 8-OHdG levels. We will then determine whether PM exposure or 8-OHdG levels are associated with adverse pregnancy outcomes.

DISCUSSION

Epidemiological studies in Europe and the United States have shown that exposure to air pollution during pregnancy is associated with adverse pregnancy outcomes [19–21]. With China's rapid development, air pollution has become more serious than before, especially in Beijing [22]. The monthly average PM_{2.5} concentration is 127.1 $\mu\text{g}/\text{m}^3$ in December 2016 in Beijing, which is higher than the current U.S. EPA National Ambient Air Quality Standard (NAAQS) of 12 $\mu\text{g}/\text{m}^3$. However, only a few studies have directly linked PM_{2.5} to pregnancy

outcomes in China. It is not clear whether premature birth, low birth weight, stillbirth, congenital anomalies, other adverse maternal outcomes, or infant respiratory or circulatory issues are significantly associated with environment exposure. This study will establish a birth cohort in Beijing and idetermine whether there are any associations between prenatal exposure to air pollution and adverse pregnancy outcomes. This work will explore the mechanisms underlying the impact of early-life exposure on adverse pregnancy outcomes. Particulate matter air pollution exposure is a known risk factor for the induction of inflammatory and oxidative stress response during pregnancy [23]. Maternal exposure to air pollutant might exacerbate oxidative damage and generate reactive oxygen species (ROS) that would induce DNA damage. Most studies indicate 8-OHdG is a biomarker for oxidative stress, and exposure to air pollution early in life may play a role in increasing systemic oxidative stress, at level of 8-OHdG in both mother and child[24,25]. We will measure the 8-OHdG levels of mother and babies to explore the relationship between exposure to air pollutants during different time windows in pregnancy and oxidative stress. We plan to determine whether air pollutants exposure or 8-OHdG levels are associate with adverse pregnancy outcomes and infant respiratory and cardiovascular diseases. We will attempt to explore the mechanisms underlying these adverse outcomes caused by air pollutant factors.

Ethical Approval

The protocol has been approved by the Medical Ethics Committee of Beijing Obstetrics and Gynecology Hospital, Capital Medical University. Written informed consent will be obtained from all study participants prior to enrolment in the study.

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Contributors

YC is the chief designer of this trial. JS, LW, YM, NT, SYH, YMD, LHZ and YYK are coinvestigators. YMD assisted in designing the study and developed the objectives and methods of the study. JS and LW will draft the manuscript, and YC is responsible for supervision of the study and revision of the manuscript. All authors have contributed to the composition of the study protocol and have approved the final version of this manuscript. YC, JS, LW, YM, and NT are involved in the recruitment of patients. SYH is responsible for the collection of clinical data for pregnant women. LHZ is responsible for the management of biological sample library. YYK is responsible for neonatal follow-up.

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Data Sharing Statement

The findings of this study will be disseminated via the usual scientific forums including peer-reviewed publications and presentation at international conferences.

Competing Interests

None declared.

Trail Registration Number

This study has been registered in WHO International Clinical Trial Register- Chinese Clinical Trial Registry under registration number ChiCTR-ROC-16010181. (<http://www.chictr.org.cn/showproj.aspx?proj=17328>)

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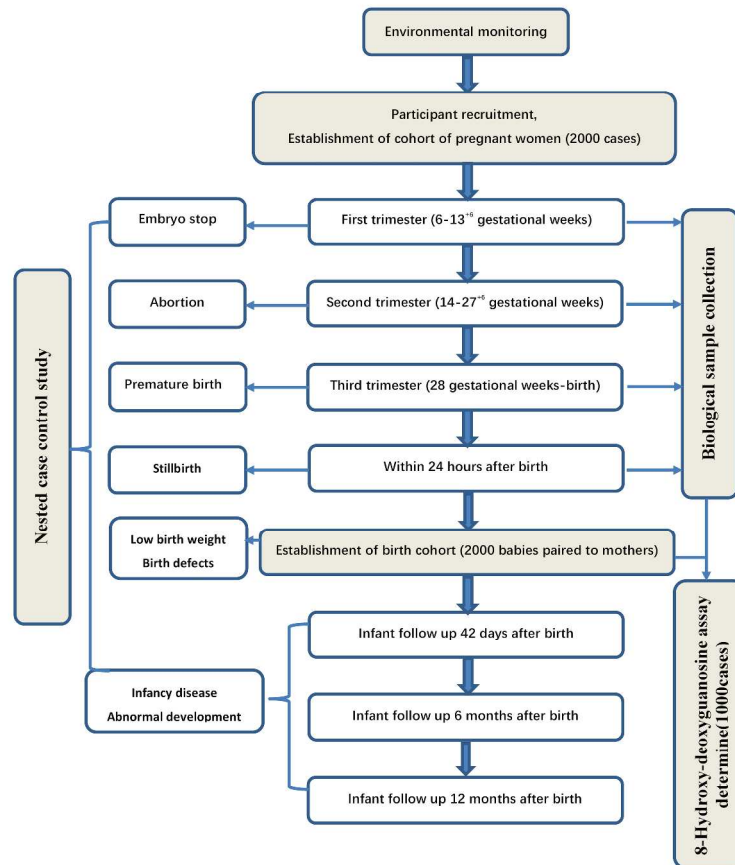
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Figure 1: flow chart

For peer review only



flow chart

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Early Life Exposure to Air Pollutants and adverse pregnancy outcomes: survey protocol in Beijing

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**Early Life Exposure to Air Pollutants and adverse pregnancy outcomes:
survey protocol in Beijing**

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ABSTRACT

Introduction: The association between early exposure to ambient air pollution and adverse pregnancy outcomes in China is unclear. This study will assess the risk of early life exposure to air pollutants in Beijing and explore the viability of 8-hydroxy-deoxyguanosine as a biological indicator to assess oxidative stress induced by early life exposure to air pollution.

Methods and analysis: Here, 2500 women with singleton pregnancies and their infants will be recruited from the Beijing Obstetrics and Gynecology Hospital. We will collect 9 types of biological samples, including maternal serum, urine, placental tissue, umbilical cord tissue, and umbilical cord blood during all three trimesters. The air pollution data (PM2.5, PM10, and similar factors) will be recorded at official fixed-site monitoring stations closest to where the pregnant women live. We plan to assess the effect of air pollutants on adverse pregnancy outcomes and infant respiratory and circulatory disease using Cox regression and competitive risk analysis and explore possible critical windows of exposure during pregnancy using daily pollutant concentrations averaged over various periods of pregnancy combined with individual activity and physiological parameters. Maternal and umbilical cord blood samples (1000 samples)

will be randomly selected for 8-hydroxy-deoxyguanosine assays to assess the correlation between exposures to air pollutants and oxidative stress. We will determine whether air pollutant exposure or 8-OHdG levels are associated with adverse pregnancy outcomes. SPSS and SAS statistical software will be used for data analysis. Cox regression and competing risk analysis will be used to compute the hazard ratio and population attributable risk.

Ethics and dissemination: This research protocol has already been approved by the Medical Ethics Committee of Beijing Obstetrics and Gynecology Hospital. Written informed consent will be obtained from all study participants prior to enrollment. The results will be published in peer-reviewed journals or disseminated through conference presentations.

Strengths and limitations

1. This is the first birth cohort study to focus on the early-life exposure to air pollutants and adverse pregnancy outcomes in Beijing.
2. This research aims to further investigate the mechanisms underlying the impact of air pollution to adverse birth outcomes.
3. This study should provide a reference for future large-scale studies of early life exposure to air pollutants in China.
4. Restriction to a single city limits the generalizability of our research because the proportion of PM2.5 from regional and local sources varies by geographic location.
5. In this study, we consider age, socioeconomic predictors, daily activities and time spent indoors confounding factors. However, individual-level exposure will not be quantified by personal sampling. Quantification of an individual’s exposure is imprecise without either a completely controlled environment or portable sampling equipment.

INTRODUCTION

Maternal and child health is the premise and basis of healthy development; it is also related to the happiness and well-being of millions of households. Thus, people are making every effort to reduce the incidence of adverse pregnancy outcomes and neonatal morbidity. The rapid economic development and urbanization of China have led to worsening trends in the overall quality of urban and regional environments [1,2]. The impact of air pollution on human health has long attracted attention. Air pollution is not only directly linked to elevated mortality and morbidity through respiratory and cardiovascular diseases, but also leads to adverse pregnancy outcomes [3-6]. It not only affects infants directly but it can also have a long-term impact on adult health, including inducing hypertension, coronary heart disease, and non-insulin-dependent diabetes [7]. The impact of air pollution on adverse pregnancy outcomes has been studied internationally for some time, and the epidemiological studies include cross-sectional surveys, follow-ups and cohort studies [8-10], which have strong abilities to demonstrate a causal relationship. Time series and case-crossovers have been used to study the impact of air pollution (PM_{2.5}, PM₁₀, SO₂, NO₂ and O₃) on adverse pregnancy outcomes such as low birth weight, premature birth, birth defects, stillbirth and intrauterine inflammation, and maternal conditions such as preeclampsia and diabetes mellitus [11-16]. In recent years, with the development of technology and cross-disciplinary approaches, individual maternal exposure to air pollutants is estimated by combining data from environmental monitoring stations and the nearby residential addresses of the mothers in question. An increasing number of studies have examined the effects of air pollution on adverse pregnancy outcomes, including studies in China [17,18]. According to the World Health Organization air pollution database, China has higher levels of air pollution than Western countries. However, research into the effect of air pollution on adverse pregnancy outcomes in China has lagged and is quite limited. Environmental pollution studies have focused

mainly on persistent organic pollutants, heavy metal pollution, and atmospheric pollution. Most studies have used multivariate logistic regression or time series to analyze the relationship between exposure to atmospheric PM10, SO₂, and NO₂ during pregnancy and adverse pregnancy outcomes. The main adverse pregnancy outcomes studied have been preterm birth, birth defects, and low birth weight. China's national environmental protection bureau monitored PM2.5 in 2013. There are only a few studies directly linking PM2.5 to long-term health effects in China. Given the large number of births in China, the impact of air pollution on adverse pregnancy outcomes must not be disregarded. However, studies on this issue in China are relatively limited. It therefore is necessary to conduct large-sample prospective cohort studies to quantitatively evaluate individual exposure and to analyze the impact of air pollution on birth outcomes and explore the biological mechanisms.

OBJECTIVES

Primary Objectives

The primary aims of this study are to establish a maternal and birth cohort, then set up a maternal and newborn information database and biological sample library based on that cohort.

We will record the air pollutant indicators published daily by the Environmental Monitoring Station near the pregnant women's residences, including PM2.5, PM10, SO₂, and NO₂ at different stages of pregnancy.

The purpose of this work is to study the correlations between environmental exposure early in life and adverse pregnancy outcomes and to investigate the correlations between different levels of exposure to ambient air pollution at different stages of pregnancy and adverse pregnancy outcomes including preterm birth, fetal growth restriction, low birth weight, and the

correlations to birth defects like congenital heart disease, neural tube defects, cleft lip and palate, and similar conditions.

Secondary Objectives

To explore the significance of using 8-hydroxy-deoxyguanosine as a biological indicator to assess early life exposure to air pollutants and investigate the mechanisms underlying these adverse outcomes.

STUDY DESIGN

Study Population and Cohort Design

This is a prospective cohort study. A total of 2500 singleton pregnant women will be recruited from Jan. 1, 2017 to Dec. 31, 2018 during their routine antenatal check at the Beijing Obstetrics and Gynecology Hospital of Capital Medical University. They will undergo delivery in the hospital and must plan to reside in Beijing for at least the next 3 years. They must provide informed consent and agree for their newborns to participate in the birth cohort for a 1-year follow up. This research is performed to establish a maternal and birth cohort and biological sample library studying covering correlations between exposure to air pollutants early in life and adverse pregnancy outcomes. A flow chart of this trail is given in Figure 1.

Exclusion Criteria

Each patient's history of contact with radioactive material during early pregnancy, history of use of any teratogenic drug or fever during early pregnancy (body temperature $>38.5^{\circ}\text{C}$); family history of genetic disease; inbreeding; and chromosome abnormalities of pregnant women or their husbands are to be recorded.

Data Collection and Management

1. Estimated sample size

We refer to incidence of adverse outcome and RR value under exposure reported in the literature and use this formula to calculate the sample size. The estimated sample size is no more than 2500 cases with 90% statistical power.

$$n = \frac{(Z_{\alpha} \sqrt{2pq} + Z_{\beta} \sqrt{p_0q_0 + p_1q_1})^2}{(p_1 - p_0)^2}$$

2. Collection of participant data

Pregnant women will be recruited according to the inclusion criteria. After providing written informed consent, they will fill out the questionnaires with the help of trained investigators. The maternal population’s financial information, detailed residential history going back three months before pregnancy and through early pregnancy, health condition, lifestyle, history of pregnancy, anamnesis, mode of conception, and condition during early pregnancy will be noted. Detailed individual information collected from pregnant women (through a questionnaire, detailed in Questionnaires): Age, height, weight, body mass index, occupation (occupational exposure), education, and other demographic data. Detailed residential address information, daily outdoor activities and time spent outdoors, outdoor exertion and protective measures used during smog episodes, usage of air purifying apparatus, and cooking habits. Health condition before pregnancy, menstrual history, anamnesis and pregnancy history, mode of conception, history of smoking or passive smoking, alcohol consumption, lifestyle during the three months preceding pregnancy, history of adverse environmental exposure, dietary habits, folic acid intake, morning sickness, mental health, and some information regarding the husband,

including height, weight, occupation, smoking, and alcohol assumption will be recorded.

Everyone's information will be entered into the database and numbered, and each infant will be assigned a number matching his or her mother's. The pregnant women will enter the cohort early during pregnancy and be followed throughout early pregnancy (6–13⁺⁶ gestational weeks), the second trimester (14–27⁺⁶ weeks), the third trimester (> 28 weeks), birth, 42 days after birth, and the neonatal and infant stages; information including clinical data such as clinical phenotype, laboratory results, ultrasound images, treatment and prognosis, and infant growth data will be systematically collected. Newborns will enter the birth cohort upon birth. A medical database with 2500 pairs of mothers and infants covering the entire process through pregnancy and infancy will be established. Outcome measurements will be checked by a clinical expert. These data will be written on the case report form (CRF) by a certificated clinical research coordinator.

3. Biological sample collection

Biological samples will be collected based on the establishment of the maternal and birth cohorts. Biological sample libraries containing maternal serum, urine collected during the 1st trimester (1–13⁺⁶ weeks), 2nd trimester (14–27⁺⁶ weeks), 3rd trimester (28 weeks to delivery), and at birth will be established. Placental tissue, umbilical cord tissue, umbilical cord blood, meconium, children's dried blood spots, maternal hair, and colostrum/breast milk will be collected after delivery. CVS samples and amniotic fluid samples (10 ml) will be collected from cases who receive prenatal diagnosis of fetal malformations found. A total of more than 35,000 samples are expected.

Maternal peripheral venous blood (5 ml) and urine (5 ml) will be collected during early pregnancy, the second trimester, the third trimester, and 24 h after birth. Serum and urine will be separated and cryopreserved at –80°C. CVS samples and amniotic fluid samples (10 ml) will be

collected from cases who received a prenatal diagnosis of fetal malformations during the second trimester.

Umbilical cord blood (5 ml), the umbilical cord itself (10 cm), and 2×2×2 cm full-thickness placental samples (rinsed in ice-cold saline to remove blood) will be collected and cryopreserved at −80°C at the time of birth. Then 5 ml of maternal urine, a strand of hair 3 mm in length and maternal colostrum/breast milk (1 ml), will be collected 24 h after birth and cryopreserved in the library. The delivery process and newborn’s condition will be recorded in detail, and the survey questionnaire will be completed by the physician. A detailed birth index will be recorded. It will include gestational week, infant gender, weight, length, Apgar score, neonatal complications, and appearance and organ abnormalities. Each baby’s meconium will also be collected to examine metabolites and assess the fecal microbiota. Dried blood spot specimens in one heel of each newborn will be collected onto specially manufactured absorbent filter paper and stored in plastic bags.

4. Infancy visits

An early growth monitoring system will be established based on the maternal and birth cohort and newborns will be included starting at birth. Anthropometric measurements and questionnaires related to feeding and sickness will be conducted at a community health care center when the children are 42 days, 6 months, and 12 months old. At 42 days after birth, the mother will bring the newborn to the health center. Information pertaining to infant growth, development, and breastfeeding will be recorded, and the survey questionnaire will be completed by the physician. At 6 and 12 months after birth, the infant will be brought to the community health care clinic for anthropometric measurements and evaluation of its development. Infant development will be monitored, and height, weight, head circumference, and number of teeth

will be determined, and questionnaires will be completed regarding feeding and illnesses, including diseases of the respiratory or cardiovascular system as diagnosed by doctors.

5. Air pollution data collection for exposure assessment

There are 35 air quality monitoring stations distributed in the different districts in Beijing which is in line with national air quality monitoring position with the requirement of technical specification. The air pollution level and population exposure will be evaluated using environmental monitoring data (PM_{2.5}, PM₁₀, SO₂, NO₂, and O₃) published daily by the Beijing Municipal Environmental Monitoring Center. Data will be collected from the closest monitoring station where pregnant women live. Daily meteorological data such as temperature and humidity will also be collected from the national meteorological center and used as a confounding control in the analysis. We plan to explore potential critical windows of exposure during pregnancy using daily PM_{2.5}, PM₁₀, and other pollutant concentrations averaged over various periods, including the three months before pregnancy and each of the three trimesters of pregnancy. Exposure throughout the entire pregnancy will be calculated as the mean of all days of the pregnancy.

6. Recording of outcome measures

Adverse pregnancy outcomes include birth defects, preterm birth, low birth weight, and fetal growth restriction. Associated birth defects include anencephaly, neural tube defects (anencephaly, spina bifida, congenital spina bifida, congenital hydrocephalus), cleft lip and palate (cleft palate, cleft lip, cleft lip with cleft palate), ear deformity (microtia including anotia, outer ear and other malformations), gastrointestinal abnormalities (esophageal atresia or stenosis, anorectal atresia or stenosis (including no anus), urinary tract malformations (hypospadias, bladder exstrophy), abdominal malformation (congenital diaphragmatic hernia, omphalocele,

gastroschisis), Down syndrome (trisomy 21), congenital heart disease, limb malformations (varus clubfoot, polydactyly, syndactyly and longitudinal deficiency), limb shortening, and ambiguous genitalia. Preterm birth refers to birth at a gestational age of 28 to 36⁺⁶ weeks (gestational age is calculated according to the last menstrual period combined with an early pregnancy ultrasound image). Low birth weight refers to birth weight below 2500 g.

Data Analysis

1. Investigate the relationship between adverse air pollutant exposure of the population and offspring development outcomes (specifically adverse pregnancy outcomes)

Data will be analyzed to evaluate the association between air pollutant exposures and pregnancy outcomes. All study participants will be closely followed up throughout the pregnancy, and pregnancy complications (such as pregnancy-induced hypertension, gestational diabetes, fetal growth restriction, premature rupture of membranes, and intrauterine infection) will be recorded in detail in a timely manner. Based on PM2.5, PM10, SO₂, NO₂, and O₃ concentrations and on many kinds of calculation methods, such as the closest point method and inverse distance weighting, combined with the individual activity patterns and physiological parameters, we will calculate the hazard ratio and population attributable risk using Cox regression and competitive risk analysis in order to assess the effect of air pollutants on adverse pregnancy outcomes.

2. Investigate the relationship between early life exposure and oxidative stress in pregnant women and fetuses

Studies showed that exposure to particulate air pollution early in life plays a role in increasing systemic oxidative stress. Oxidative stress is believed to be involved in the mechanisms underlying the effects of air pollution on human health. 8-hydroxy-2'-deoxy

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4 guanosine (8-OHdG) is a sensitive indicator of oxidative stress. We will investigate 8-OHdG
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6 levels during different time windows in maternal blood and cord blood and explore the
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8 relationship between exposure to air pollutants and oxidative stress. Maternal and umbilical cord
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10 blood samples (1000 each) will be randomly selected from the cohort for
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12 8-hydroxy-deoxyguanosine assay to determine the correlation of early life exposures to oxidative
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14 stress in pregnant women and fetuses. All recruited pregnant women paired with their babies will
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16 be numbered starting at 1, and then either odds or evens will be randomly selected for testing.
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18 19 20 **Quality Control**

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22 Data management: The questionnaires were designed based on extensive reviews of the
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24 literature and advice from obstetrics and pediatrics experts. Detailed standards will be
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26 established for subject recruitment of the maternal and birth cohort. The exact addresses of the
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28 pregnant women will be recorded to allow selection of the closest environmental monitoring
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30 stations. Each data monitoring center is in Beijing Obstetrics and Gynecology Hospital. The
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32 participants may drop out of the trial at any time for any reason. The physician should assess
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34 each patient and provide any needed assistance. Incidences of loss to follow-up will be reported
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36 to the chief designer of this trial weekly.
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41 Investigator training: A research manual (CRF tables) will be prepared to standardize the
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43 investigation and survey techniques and to improve the working ability and responsibility of
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45 investigators; specialized physicians will be responsible for the clinical diagnostic reports.
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48 Questionnaire verification: Clinical physicians will check the questionnaires after every
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50 survey for logical errors and missed information, and additional surveys will be conducted within
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52 24 h to supplement the data.
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55 Data entry: A combination of hard copy questionnaires and computer database entries will
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57 be used for data collection. Concurrent validity data will be collected for quality control.
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Training will be provided to all personnel involved in data entry beforehand. Two specialized personnel will be responsible for the second round of quality control assessment every month.

Development of a protocol for biological sample collection: Sample collection will be performed in strict accordance with protocol, and the rate of loss to follow-up will be closely controlled. Biological sample libraries will be managed by assigned individuals, SOP regulations will be developed, and sample collection site size, standing time, transportation, and storage will be strictly controlled according to the protocol. To ensure data accuracy, completeness, and clarity, all samples will be stored using the sample number matching the maternal information and recorded and managed via computer management software system, which will be logged with the hospital HIS system and regularly exported for storage.

Adverse events

All adverse events in any pregnant woman or infant will be recorded in detail and closely monitored until stabilization or resolution. The chief designer of this trial will be informed immediately and responsible for reporting to Scientific Research Department and Medical Ethics Committee of Beijing Obstetrics and Gynecology Hospital. Regular monitoring will be used to ensure that all adverse events are identified and addressed appropriately.

Statistical analysis

Epidata 3.1 software will be used for double-entry of the data and logic verification to establish a database; SPSS version 18.0 (SPSS, Chicago, IL, U.S.) and SAS version 9.3(SAS Institute Inc., Cary, NC, U.S.) statistical software will be used for data analysis. Descriptive analysis will be performed first to compute the general maternal and infant demographic data and distribution of birth outcomes and birth defects. Cox regression and competing risk analysis will be used to compute the hazard ratio (HR) and population attributable risk (PAR) and to estimate

the effect of main air pollutants on major adverse birth outcomes (low birth weight, preterm birth and fetal growth restriction) and birth defects (congenital heart disease). Pearson correlation coefficients and multiple linear regression will be used to assess the association between air pollutant exposure and 8-OHdG levels. We will then determine whether PM exposure or 8-OHdG levels are associated with adverse pregnancy outcomes.

DISCUSSION

Epidemiological studies in Europe and the United States have shown that exposure to air pollution during pregnancy is associated with adverse pregnancy outcomes [19–21]. With China's rapid development, air pollution has become more serious than in previous years, especially in Beijing [22]. The average PM_{2.5} concentration hit 127.1 µg/m³ in December 2016 in Beijing, which is higher than the current U.S. EPA National Ambient Air Quality Standard (NAAQS) of 12 µg/m³. However, only a few studies have directly linked PM_{2.5} to pregnancy outcomes in China. It is not clear whether premature birth, low birth weight, congenital anomalies, other adverse maternal outcomes, or infant respiratory or circulatory issues are significantly associated with environment exposure. This study will establish a birth cohort in Beijing and determine whether there are any associations between prenatal exposure to air pollution and adverse pregnancy outcomes. This work will explore the mechanisms underlying the impact of early-life exposure on adverse pregnancy outcomes. Particulate matter air pollution exposure is a known risk factor for the induction of inflammatory and oxidative stress response during pregnancy [23]. Maternal exposure to air pollutants might exacerbate oxidative damage and generate reactive oxygen species (ROS) that would induce DNA damage. Most studies indicate 8-OHdG is a biomarker for oxidative stress, and exposure to air pollution early in life

may play a role in increasing systemic oxidative stress, at levels of 8-OHdG in both mother and child [24, 25]. We will measure the 8-OHdG levels of the mothers and babies to explore the relationship between exposure to air pollutants during different time windows in pregnancy and oxidative stress. We plan to determine whether 8-OHdG levels or exposure to air pollutants are associated with adverse pregnancy outcomes and infant respiratory and cardiovascular diseases. We will attempt to determine the mechanisms underlying these adverse outcomes caused by air pollutant factors.

Ethical Approval

The protocol has been approved by the Medical Ethics Committee of Beijing Obstetrics and Gynecology Hospital, Capital Medical University. Written informed consent will be obtained from all study participants prior to enrollment in the study.

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Contributors

YC is the chief designer of this trial. JS, LW, YM, NT, SYH, YMD, LHZ and YYK are coinvestigators. YMD assisted in designing the study and developed the objectives and methods of the study. JS and LW will draft the manuscript, and YC is responsible for supervision of the study and revision of the manuscript. All authors have contributed to the composition of the study protocol and have approved the final version of this manuscript. YC, JS, LW, YM, and NT are involved in the recruitment of patients. SYH is responsible for the collection of clinical data

for pregnant women. LHZ is responsible for the management of biological sample library. YYK is responsible for neonatal follow-up.

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Data Sharing Statement

The findings of this study will be disseminated via the usual scientific forums including peer-reviewed publications and presentation at international conferences.

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Competing Interests

None declared.

Trial Registration Number

This study has been registered in WHO International Clinical Trial Register- Chinese Clinical Trial Registry under registration number ChiCTR-ROC-16010181.
(<http://www.chictr.org.cn/showproj.aspx?proj=17328>)

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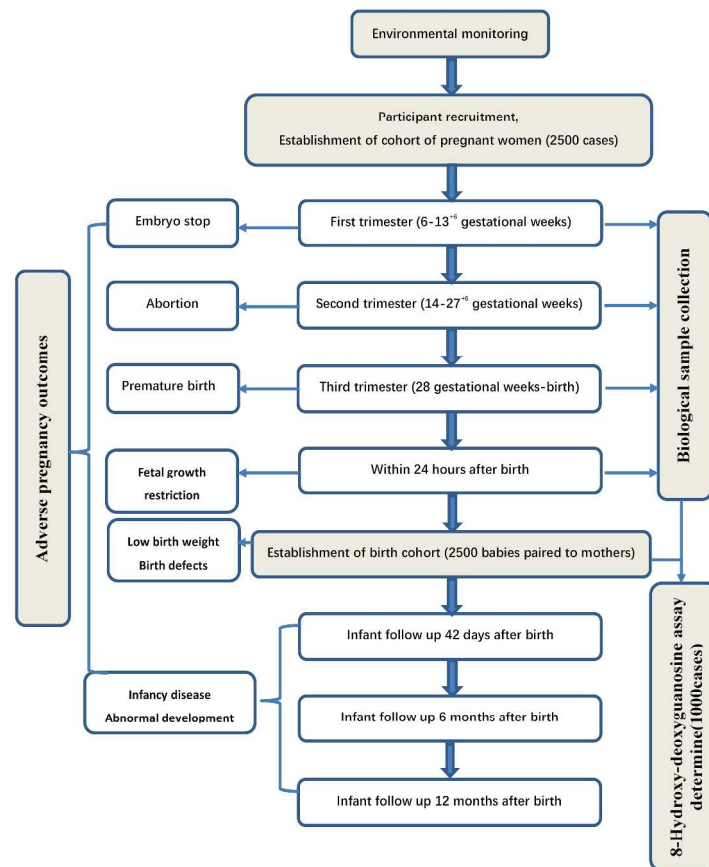
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Figure 1: Flow chart

For peer review only



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